Einstein’s equation to provide humanity with an essentially limitless supply of energy.

In these examples, energy is produced from mass. But Einstein’s equation works perfectly well in reverse—the direction in which mass is produced from energy—and that’s the direction in which string theory uses Einstein’s equation. The *mass* of a particle in string theory is nothing but the *energy* of its vibrating string. For instance, the explanation string theory offers for why one particle is heavier than another is that the string constituting the heavier particle is vibrating faster and more furiously than the string constituting the lighter particle. Faster and more furious vibration means higher energy, and higher energy translates, via Einstein’s equation, into greater mass. Conversely, the lighter a particle is, the slower and less frenetic is the corresponding string vibration; a massless particle like a photon or a graviton corresponds to a string executing the most placid and gentle vibrational pattern that it possibly can.[34][14]

Other properties of a particle, such as its electric charge and its spin, are encoded through more subtle features of the string’s vibrations. Compared with mass, these features are harder to describe nonmathematically, but they follow the same basic idea: the vibrational pattern is the particle’s fingerprint; all the properties that we use to distinguish one particle from another are determined by the vibrational pattern of the particle’s string.

In the early 1970s, when physicists analyzed the vibrational patterns arising in the first incarnation of string theory—the *bosonic string theory* — to determine the kinds of particle properties the theory predicted, they hit a snag. Every vibrational pattern in the bosonic string theory had a whole-number amount of spin: spin-0, spin-1, spin-2, and so on. This was a problem, because although the messenger particles have spin values of this sort, particles of matter (like electrons and quarks) don’t. They have a fractional amount of spin, spin-½. In 1971, Pierre Ramond of the University of Florida set out to remedy this deficiency; in short order, he found a way to modify the equations of the bosonic string theory to allow for half-integer vibrational patterns as well.